Staff answers · shared work

Starter

The margins show possible gamete alleles; each internal box combines one from each parent.

Worked example

Construct BB × bb in a simple one-gene model.

A BB parent can contribute only B gametes.

A bb parent can contribute only b gametes.

Combine one from each: every predicted offspring is Bb.

The predicted chance of Bb is 100% under these model assumptions.

Shared evidence task

Aa × Aa gives AA, Aa, Aa, aa: aa probability 1/4. Aa × aa gives Aa, Aa, aa, aa in an equally weighted 2×2 representation: aa probability 1/2.

Card placements

Card	Group
A	Gamete
a	Gamete
AA	Offspring pair
Aa	Offspring pair
aa	Offspring pair
Aa in one gamete box	Wrong notation

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Staff answers · Supported

Arrival

1. Pair: The two alleles differ.

2. Gamete: One.

3. Options: A or a.

4. Combine: Aa.

Independent / application

1. Construct: Margins A,a and A,a; boxes AA,Aa,Aa,aa.

2. Compare: Aa,Aa,aa,aa; probability aa = 2/4 = 1/2 = 50%.

3. Explain: A from one parent and a from the other can combine in two ways, both producing Aa.

Two-check exit

1. Construct: AA, Aa, Aa, aa.

2. Interpret: 2/4 = 1/2.

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Staff answers · Independent

Arrival

1. Pair: The two alleles differ.

2. Gamete: One.

3. Options: A or a.

4. Combine: Aa.

Independent / application

1. Construct: Margins A,a and A,a; boxes AA,Aa,Aa,aa.

2. Compare: Aa,Aa,aa,aa; probability aa = 2/4 = 1/2 = 50%.

3. Explain: A from one parent and a from the other can combine in two ways, both producing Aa.

Two-check exit

1. Construct: AA, Aa, Aa, aa.

2. Interpret: 2/4 = 1/2.

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Staff answers · Aspire

Arrival

1. Pair: The two alleles differ.

2. Gamete: One.

3. Options: A or a.

4. Combine: Aa. Accept a scientifically correct reason that uses the relevant model or evidence.

Independent / application

1. Construct: Margins A,a and A,a; boxes AA,Aa,Aa,aa.

2. Compare: Aa,Aa,aa,aa; probability aa = 2/4 = 1/2 = 50%.

3. Critique a square: An Aa parent makes gametes with A or a. Put A and a in each margin, then combine to make diploid offspring genotypes.

Two-check exit

1. Construct: AA, Aa, Aa, aa.

2. Critique a square: An Aa parent makes gametes with A or a. Put A and a in each margin, then combine to make diploid offspring genotypes.

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Staff · source and teaching care

Weekly binding: Complete a genetic cross (digital Punnett square).

Exact pupil goal: Construct a monohybrid Punnett square and explain how gametes combine to predict offspring genotypes.

Source: Launch/W8-W13_2026-27/SCI_L_W13L2_Punnett_Square_Explore.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d.

Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B43 Aut2·W5; C43 Complete a genetic cross (digital Punnett square).

Sources are inherited from the verified lesson. No new specification alignment is asserted. [AWARD] [QUAL] names remain masked. No biological or personal family data is used.

Hearing both pupils matters. A correct click is not evidence of reasoning. Re-model the uncertain step for one pupil while the other attempts a fresh example. Preserve the original answer and the prompt used.

A genetic counsellor explains chance without promising a fixed outcome. This is a career link, not advice about a person.

Model care: This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

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